

Md. Asadujjaman Miah

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Field & Programming Skills

Field: Optimization, numerical methods, Machine learning, Neural networks, Mathematical modeling, PINNs,

Programming: Python, MATLAB, COMSOL, FORTRAN, Mathematica

Research Experience

1. Intern at OIST “Comparing Individual-Based Deterministic and Stochastic Models in Moment-Based Dynamics”
2. Comparing Physics-Informed RaNNs and Physics-Informed Neural Networks ”Remote contributor Numerical Analysis group TU Delft ”
3. Solving stochastic and normal Volterra IEs of the discontinuous kernel problem using a neural network under Dr Samad Noeiaghdam (Google Scholar) (January 2025 – Present)
4. Designing efficient numerical root-finding algorithm for solving transcendental equations under Md. Shorif Hossan (Google Scholar) (October 2023 – November 2024)
5. Final year academic project: A study of meteorological parameters behavior over the Rajshahi region using standard statistical and numerical methods under Dr. MA Samad (Google Scholar) (January 2025 – Present)

Publications & Presentations

Published

- ”Derivative-Free Root-Finding Algorithms: Perpendicular and Extended Secant Methods with CESTAC Validation” (Elsevier’s Q1 journal of Applied Mathematics and Computation) 10.1016/j.amc.2026.130053
- Main Project published in *Axioms* (MDPI), indexed in Scopus and Web of Science (Q2 in Applied Mathematics)
<https://www.mdpi.com/2075-1680/14/10/726>

Conference Presentation

- “Homotopy Analysis Method and Physics-Informed Neural Networks for Solving Volterra Integral Equations with Discontinuous Kernels”, 7th International Conference DYSC 2025, Irkutsk, Russia Link)
- ”Novel Algorithms for Nonlinear Root-Finding: Perpendicular and Extended Secant Methods with CADNA Implementation” (Accepted for Oral presentation at GAMM 2026 Annual meeting)

Accepted

- “Hybrid HAM-Neural Network Framework for Nonlinear Functional Hammerstein Integral Equations with CESTAC Validation” (*Journal of Integral Equations and Applications*)

Under Review

- “Regularization and Neural Correction in the Homotopy Analysis Framework for First Kind Volterra Integral Equations with Discontinuous Kernel” (*Bulletin of Irkutsk State University, Series Mathematics*)
- “Precision-Guaranteed Computation for Functional Volterra Integral Equations: Integrating ADM, Aitken Acceleration, and Stochastic Validation” (*Journal of Computational Mathematics*)
- “Fractional order supply-Demand model by PINNs”
- “Statistical Assessment of Wind Energy Potential in Bangladesh: Weibull and Rayleigh Distribution Analysis”
- “Stochastic Analysis of Control Parameter Influences in an Eco-Epidemiological Model with Predator Infection” (*Scientific Reports*)

- “Predicting risky areas due to subsidence using the ensemble-based Machine Learning method in the Barapukuria coal mine” (*Mining, Metallurgy & Exploration*)

Project on NN and Sci-ML

- Generating synthetic data from a given function by applying noise and using scientific machine learning to find ground truth
GitHub Code

Certifications

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI, Stanford University (Certificate Link)
- Intro to Machine Learning – Kaggle (Certificate)
- Advanced Learning Algorithms – DeepLearning.AI & Stanford University (Certificate Link)

Awards

2nd Position, Undergraduate Level International Day of Mathematics Olympiad

Education

B.Sc in Applied Mathematics, University of Dhaka, Bangladesh January 2022 – Present
Current CGPA: 3.68

Languages

English (Professional), Bangla (Native)